

Fraud Detection & AML Alert Triage

Client: STADIOEquities

Part C : Data Request

Purpose of this request:

This request specifies the data STADIOEquities needs to provide in order to investigate whether compliance alerts generated by the existing rules-based transaction monitoring system can be reliably scored by likelihood of being a genuine case versus a false positive. As mentioned in the Problem Statement, the compliance monitoring system already generates transaction-monitoring alerts, dispositions and case outcomes; this data forms the core labelled dataset the model will be trained on. Six supporting tables (account and funding history, client demographics, transaction history, app/web session behaviour, support tickets and public app-store reviews) provide the features and corroborating signals needed to explain why an alert was raised and to distinguish genuine risk from routine, explainable client behaviour.

Tables 6 and 7 are included specifically because the Motivation for this project argues that unresolved false positives are linked to a rise in withdrawal-related support tickets and plausibly, to public complaints that affect prospective clients during due diligence. Table 6 (support tickets) lets the support-ticket link be tested directly at the account level; Table 7 (app-store reviews) test the public facing side of that argument at an aggregate level, since reviews cannot be linked to individual accounts.

In line with the guidance to request more detail than may initially seem necessary, each table has been scoped generously so that if there are potential data quality or completeness issues they can be identified and resolved before modelling begins rather than discovering these issues midway through the project.

Extraction period: please provide the full available history for each table rather than a narrow recent window - up to four years for compliance, account and funding, transaction, and app/web session data, and up to three years for support and feedback data, per the history currently available in each system. Please extract through to the most recent data available. Longer history allows more reliable client level behavioural baselines to be built.

Note on data privacy: No column in this request asks for a client's name. ID/passport number, physical address, contact details or full device serial number. Where a field could reasonably reveal a client or staff member's identity (e.g. investigator notes, device identifiers, support ticket text). The request specifies that it be provided in anonymised or masked form. All identifiers below are intended to be internal system generated ID's, not personal details.

Table 1: Compliance Monitoring – Alerts and Case Outcomes

This is the core table for the project. It contains historical record of alerts raised by the rules engine together with the investigators final finding, which supplies the target label (true positive / false positive) the model will learn to predict. One row per alert

Column name	Data type	Example	Additional information
Alert ID	String, categorical	ALT-2026-004821	Unique identifier for each alert generated by the rules engine. Primary key for this table. One row per alert.
Alert timestamp	String, datetime	2026-03-14 09:12:45	Date and time alert was raised, in yyyy-MM-DD hh:mm:ss format.
Rule triggered	String, categorical	Rapid deposit-withdrawal	The specific rule or threshold that generated the alert. Please provide the full list of valid rule names currently in use, e.g. "Large withdrawal", "Rapid deposit-withdrawal", "High-frequency trading", "New account high value", "Structuring pattern", "Other".
Account ID	String, categorical	ACC-0234871	Foreign key linking this alert to the account it was raised against (See Table 2).
Transaction ID	String, categorical	TXN-98217364	Foreign key linking this alert to the specific transaction that triggered it (see Table 4). Leave blank if alert was raised against account level behaviour rather than a single transaction.
Rule-assigned severity score	Numeric, integer	3	The severity or priority score assigned automatically by the rules engine at the point of generation, if such a score exists in the current system. Valid range: 1 (lowest) to 5 (highest). Leave blank if no score is generated.
Investigation disposition	String, categorical	False positive	The final outcome of the investigator's review. Valid values: "True positive", "False positive", "Unresolved". This is the target/label field the model will be trained to predict completeness of this column is critical to the project. Note: "Unresolved" cases carry no known outcome and will be excluded from model training, though they are still useful for understanding overall alert volume, so please include them rather than filtering them out.
Case closed timestamp	String, datetime	2026-03-16 14:02:10	Date and time the case was closed, in yyyy-MM-dd hh:mm:ss format. Used to calculate investigator resolution time. Leave blank if the case is still open.

Investigator notes (anonymized)	String, free text	"Client confirmed salary deposit via payslip"	Free text notes recorded by the investigator during case review. Please remove or mask any client names, ID numbers, account numbers, or other personally identifying text before providing this field.
Assigned investigator ID (anonymised)	String, categorical	INV-014	An anonymised staff identifier only (not the investigators name), included to check for potential reviewer level patterns in how alerts are dispositioned.

Table 2: Account and Funding Profile

One row per client account. Provides the account level context (age of account, funding status, KYC completeness) needed to judge whether flagged behaviour is unusual for that specific client.

Column name	Data type	Example	Additional information
Account ID	String, categorical	ACC-0234871	Unique identifier for each client account. Primary key for this table.
Registration date	String, datetime	2025-11-02	Date the account was first registered, in yyyy-MM-dd format.
KYC status	String, categorical	Complete	Current state of the Know Your Customer verification process. Valid values "Complete", "Incomplete", "Pending".
KYC completion date	String, datetime	2025-11-05	Date KYC was fully verified, in yyyy-MM-dd format. Leave blank if KYC status is not "Complete".
First deposit date	String, datetime	2025-11-06	Date of the clients first successful deposit, in yyyy-MM-dd format. Leave blank if the account has never been funded.
Account status	String, categorical	Active	Current status of the account. Valid values: "Active", "Dormant", "Closed".
Premium subscription status	String, categorical (Boolean)	No	Whether the account currently holds a premium subscription tier. Valid values: "Yes" or "NO".

Table 3: Client Demographics

One row per client account. Supplies the demographic, stated-intent, and stated risk-appetite context available at sign up, used to help distinguish plausible client behaviour from anomalous behaviour.

Column name	Data type	Example	Additional information
Account ID	String, categorical	ACC-0234871	Foreign key linking to Table 2.
Age	Numeric, integer	29	Clients age in years at the time of date extraction.

Province/region	String, categorical	GP	Client stated province of residence. All nine south African provinces, i.e. "GP", "EC", "FS", "KZN", "LP", "MP", "NC", "NW", "WC". Please add an "Other" category for clients registered outside of South Africa, if applicable.
Stated investment goal	String, categorical	Retirement	The investment goal the client selected at sign-up. Please provide full list of valid goal categories currently in use on the platform.
Stated risk appetite	String, categorical	Moderate	Risk appetite the client selected at sign-up. Valid values: "Conservative", "Moderate" or "Aggressive".

Table 4: Transaction History

One row per transaction. This table provides the transaction level detail needed to build features such as transaction size relative to account history, deposit/withdrawal frequency, time between linked transactions, and portfolio concentration.

Column name	Data type	Example	Additional information
Transaction ID	String, categorical	TXN-98217364	Unique identifier for each transaction. Primary key for this table.
Account ID	String, categorical	ACC-0234871	Foreign key linking to Table 2.
Transaction timestamp	String, datetime	2026-03-14 09:10:02	Date and time the transaction was initiated, in yyyy-MM-dd hh:mm:ss format.
Transaction type	String, categorical	Withdrawal	Valid values: "Deposit", "Withdrawal", "Buy", "Sell", "Fee".
Amount	Numeric, integer	850000	Transaction amount in cents (e.g. R8,500.00 =850000), consistent with existing STADIOEquities data conventions
Channel	String, categorical	App	The platform channel used to initiate the transaction. Valid values: "App", "Web", "ETF", "Card".
Instrument (if applicable)	String, categorical	NPN.JO	The share or instrument ticker involved for "BUY/Sell" transactions only. Leave blank for deposits, withdrawals and fees
Portfolio concentration at transaction	Numeric, decimal	0.82	The proportion of the account's total portfolio values held in its single largest instrument immediately after this transaction, as a decimal between 0 and 1 (e.g. 0.82 =82% held in one instrument) A high value indicates a concentrated, higher risk position. Leave blank for deposit, withdrawal, and fee rows where it does not apply

Account balance after transaction	Numeric, integer	4620000	The account's total balance immediately after this transaction, in cents
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Table 5: App and Web Session Behaviour

One row per login session. This table is a supporting, secondary source with behavioural signals such as device change or an unexpected login location immediately before a flagged transaction are commonly informative in separating genuine risk from false positives, even though they sit outside the core transaction and compliance data.

Column name	Data type	Example	Additional information
Account ID	String, categorical	ACC-0234871	Foreign key linking to Table 2.
Session ID	String, categorical	SES-77213456	Unique identifier for each login session. Primary key for this table.
Session start timestamp	String, datetime	2026-03-14 09:05:11	Date and time the session began. In yyyy-MM-dd- hh:mm:ss format.
Device identifier (anonymised)	String, categorical	DEV-88231	An anonymised device fingerprint (not a physical device serial number), used to detect whether a session originated from a device not previously associated with the account.
Login region/country	String/ categorical	ZA	The country (or region, if available) the session originated from, used to detect logins from an unexpected location relative to the client's usual pattern.
Primary session action	String, categorical	Withdrawal initiated	The main action taken during the session. Please provide the full list of valid action categories currently in use, e.g. "Login only", "Deposit initiated", "Withdrawal initiated", "Trade placed", "Profile update".

Table 6: Support and Feedback

One row per support ticket. This table lets the project test directly whether unresolved false positives are associated with a rise in withdrawal-related support tickets, as argued in the Motivation, and provide a corroborating, client-reported signal alongside the investigator disposition in Table 1.

Column name	Data type	Example	Additional information
Ticket ID	String, categorical	TCK-0091273	Unique identifier for each support ticket. Primary key for this table.
Account ID	String, categorical	ACC-0234871	Foreign key linking to table 2.
Ticket created timestamp	String, datetime	2026-03-14 10:02:33	Date and time the ticket was logged. In yyyy-MM-dd mm:ss format.

Ticket reason category	String, categorical	Withdrawal delayed	The categorised reason code assigned to the ticket. Please provide the full list of valid reason categories currently in use, e.g. “Withdrawal delayed”, “Account flagged/frozen”, “Fee query”, “Offshore access query”, “Other”.
Related alert ID	String, categorical	ALT-2026-004821	Foreign key linking to Table 1 where the ticket relates to a specific compliance alert (e.g. a client querying a blocked withdrawal). Leave blank if not applicable.
Ticket description (anonymised)	String, free text	“Client asking why withdrawal has been delayed for 3 days”	Free-text description of the client’s query. Please remove or mask any client names, ID numbers, or account numbers before providing this field.
Ticket resolved timestamp	String, datetime	2026-03-16 09:40:12	Date and time the ticket was resolved or closed. In yyyy-MM-dd hh:mm:ss format. Leave blank if the ticket is still open.
Ticket outcome	String, categorical	Resolved – false alarm confirmed	The final resolution category of the ticket. Please provide the full list of valid outcome categories currently in use.

Table 7: App Store Review

One row per public app-store review. Unlike tables 1-6, reviews are public and anonymous and cannot be linked to a specific Account ID, alert, or transaction. This table therefore supports the project as an aggregate, qualitative signal (e.g. tracking how often themes such as “withdrawal”, “Frozen”, or “flagged” appear in reviews over time) rather than as a row level modelling feature, and is requested to test whether public sentiment trends move in line with alert and compliant volume.

Column name	Data type	Example	Additional information
Review ID	String, categorical	REV-0041982	Unique identifier assigned to each review at the point of extraction. Primary key for this table.
Platform	String, categorical	iOS	The app store the review was posted to. Valid values: “iOS”, “Android”.
Review date	String, datetime	2026-03-10	Date the review was posted. In yyyy-MM-dd format.
Star rating	Numeric, integer	2	The star rating given by the reviewer. Valid range: 1 to 5.
Review text	String, free text	“Withdrawal has been stuck for a week, no explanation given”	Full text of the review, as posted publicly on the app store. No additional anonymisation is required, as app-store reviews are already public and anonymous by default.

App version	String, categorical	4.12.0	The app version the reviewer was using at the time of posting, if this is captured by the app store platform. Leave blank if not available.
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